

```

1 /opt/homebrew/opt/java/libexec/openjdk.jdk/Contents
  /Home/bin/java -javaagent:/Applications/IntelliJ
  IDEA.app/Contents/lib/idea_rt.jar=54120 -Dfile.
  encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.
  stderr.encoding=UTF-8 -classpath /Users/lamvo/
  MissionCollege/CIS-044/lam-vo-cis44-java-data-
  structures-labs/out/production/lam-vo-cis44-java-
  data-structures-labs SortingTester
2 --- The Sorting Race ---
3
4 --- Testing for array size n = 1000 ---
5
6 Average Case:
7 Selection Sort: 2590084 ns
8 Insertion Sort: 1620458 ns
9 Merge Sort:      463792 ns
10
11 Best Case:
12 Selection Sort: 4678291 ns
13 Insertion Sort: 3917 ns
14 Merge Sort:      69083 ns
15
16 Worst Case:
17 Selection Sort: 259958 ns
18 Insertion Sort: 378958 ns
19 Merge Sort:      58667 ns
20
21 --- Testing for array size n = 5000 ---
22
23 Average Case:
24 Selection Sort: 5957250 ns
25 Insertion Sort: 5542083 ns
26 Merge Sort:      2693667 ns
27
28 Best Case:
29 Selection Sort: 5141750 ns
30 Insertion Sort: 144750 ns
31 Merge Sort:      1685875 ns
32
33 Worst Case:
34 Selection Sort: 6682458 ns

```

```

35 Insertion Sort: 9005417 ns
36 Merge Sort:      151125 ns
37
38 --- Testing for array size n = 10000 ---
39
40 Average Case:
41 Selection Sort: 21072750 ns
42 Insertion Sort: 4899500 ns
43 Merge Sort:      700125 ns
44
45 Best Case:
46 Selection Sort: 20125958 ns
47 Insertion Sort: 14292 ns
48 Merge Sort:      255709 ns
49
50 Worst Case:
51 Selection Sort: 24881083 ns
52 Insertion Sort: 9776125 ns
53 Merge Sort:      273083 ns
54
55 --- Testing for array size n = 25000 ---
56
57 Average Case:
58 Selection Sort: 131017292 ns
59 Insertion Sort: 31210291 ns
60 Merge Sort:      1964250 ns
61
62 Best Case:
63 Selection Sort: 121737375 ns
64 Insertion Sort: 29458 ns
65 Merge Sort:      657125 ns
66
67 Worst Case:
68 Selection Sort: 149177458 ns
69 Insertion Sort: 58378834 ns
70 Merge Sort:      709500 ns
71
72 --- Testing for array size n = 50000 ---
73
74 Average Case:
75 Selection Sort: 469142500 ns

```

```
76 Insertion Sort: 118761666 ns
77 Merge Sort:      3833625 ns
78
79 Best Case:
80 Selection Sort: 468579209 ns
81 Insertion Sort: 58459 ns
82 Merge Sort:      1388083 ns
83
84 Worst Case:
85 Selection Sort: 584137375 ns
86 Insertion Sort: 244020125 ns
87 Merge Sort:      1486375 ns
88
89 --- Testing for array size n = 100000 ---
90
91 Average Case:
92 Selection Sort: 1878101375 ns
93 Insertion Sort: 489606958 ns
94 Merge Sort:      8202958 ns
95
96 Best Case:
97 Selection Sort: 1862523584 ns
98 Insertion Sort: 117458 ns
99 Merge Sort:      2950792 ns
100
101 Worst Case:
102 Selection Sort: 2320404834 ns
103 Insertion Sort: 964778750 ns
104 Merge Sort:      3159458 ns
105
106 Process finished with exit code 0
107
```